

Project Report

AI-Based Intern Performance Predictor

1. Problem Statement:

Companies and organizations need a simple way to predict an intern's performance using task completion, task scores, and submission records. Manual evaluation can take time and may not give a clear prediction of the final performance.

2. Objective:

The objective of this project is to build an AI-Based System that:

- Analyzes intern performance data.
- Predicts the overall performance score.
- Assigns a performance level.
- Provides personalized suggestions for improvement.
- Displays the prediction through a simple Streamlit web app.

3. Dataset:

The project uses intern performance data containing information about task scores, completed tasks, and submission timing. These features are used to predict the intern's overall performance.

4. Data Preprocessing:

The data was prepared before training the model. The main steps were:

- Checked the dataset for missing values.
- Cleaned and prepared the input features.
- Converted the required values into a suitable numerical format.
- Selected relevant features for prediction.
- Split the data into training and testing sets.

5. Model Training:

A machine learning regression model was trained to predict the overall intern performance score. The model learns the relationship between task performance, completion rate, and submission behavior and uses this information to predict the final score.

6. Model Evaluation:

The model was evaluated using the available intern performance data. The system produced the following results:

- Average Task Score: 73.5
- Tasks Delivered: 7/8
- On-Time Submission Rate: 50%
- Predicted Overall Score: 72.28
- Performance Level: Average

The prediction shows that the intern's current performance is at an Average level.

7. Personalized Suggestions:

The system also provides suggestions based on the predicted performance. The main areas identified for improvement were:

- NLP
- Model Deployment
- Final Capstone

These suggestions help the intern focus on areas that can improve their overall performance.

8. Streamlit Interface:

A Streamlit web interface was created to make the performance predictor easy to use.

The user can enter the required performance information, and the application displays:

- Predicted overall score
- Performance level
- Personalized improvement suggestions

The interface provides a simple way to interact with the trained model without running the Python code manually.

9. Deployment:

The Streamlit application was deployed using Cloudflare Tunnel. This provides a public web link that can be opened in a browser.

The application was successfully tested, and the deployed interface is working correctly.

10. Results

The system successfully predicted an overall score of 72.28, with a performance level of Average.

It also identified areas where improvement is needed, making the system useful for tracking and improving intern performance.

11. Conclusion:

The AI-Based Intern Performance Predictor successfully combines machine learning with a Streamlit web interface to predict intern performance. The system uses task scores, completion, and submission behavior to generate a predicted score and performance level.

The final prediction was 72.28 (Average). The system also provides personalized suggestions, making it useful for identifying areas that need improvement.

Overall, the project demonstrates a complete AI workflow from data preparation and model training to evaluation and deployment.

12. Future Improvements

The project can be improved in the future by:

- Using a larger intern performance dataset.
- Testing more machine learning models.
- Adding more performance features.
- Improving prediction accuracy.
- Adding charts and performance history.
- Deploying the application on a permanent hosting platform.
- Adding login and individual intern profiles.

13. Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Streamlit
- Cloudflare Tunnel
- Google Colab